

LECTURE SHEET

EXERCISE-I

(Coordinates of Centre of Mass, Position Vector, Velocity and Acceleration of Centre of Mass)

LEVEL-I (MAIN)

Straight Objective Type Questions

- Particles of masses 1gm, 2gm, 3gm and 4 gm are placed at $x = 1$ cm, $x = 2$ cm, $x = 3$ cm, $x = 4$ cm respectively. Then $X_{cm} =$
 1) 1cm 2) 2cm 3) 3cm 4) 4cm
- Centre of mass of two particles with masses 2 kg and 1 kg located at (1, 0, 1) and (2, 2, 0) has the co-ordinates of
 1) $\frac{4}{3}, \frac{2}{3}, \frac{2}{3}$ 2) $\frac{2}{3}, \frac{4}{3}, \frac{2}{3}$ 3) $\frac{2}{3}, \frac{2}{3}, \frac{4}{3}$ 4) $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}$
- The distance between the centres of carbon and oxygen atoms in the carbon monoxide gas molecule is 1.13×10^{-10} m. The distance of centre of mass of the molecule relative to carbon atom is
 1) 0.48×10^{-10} m 2) 0.64×10^{-10} m 3) 0.56×10^{-10} m 4) 0.36×10^{-10} m
- Masses 1 kg, 1.5 kg, 2 kg and M kg are situated at (2, 1, 1), (1, 2, 1), (2, -2, 1) and (-1, 4, 3). If their centre of mass is situated at (1, 1, 3/2), the value of M is
 1) 1 kg 2) 2 kg 3) 1.5 kg 4) 3 kg
- The co-ordinates of centre of mass of particles of mass 10, 20 and 30 gm are (1, 1, 1) cm. The position co-ordinates of mass 40 gm which when added to the system, the position of combined centre of mass be at (0, 0, 0) are,
 1) (3/2, 3/2, 3/2) 3) (-3/2, -3/2, -3/2)
 3) (3/4, 3/4, 3/4) 4) (-3/4, -3/4, -3/4)
- Particles of masses m, 2m, 3m,..... nm grams are placed on the same line at distances l, 2l, 3l, nl cm from a fixed point. The distance of centre of mass of the particles from the fixed point in centimetre is
 1) $\frac{(2n+1)l}{3}$ 2) $\frac{l}{n+1}$ 3) $\frac{n(n^2+1)l}{2}$ 4) $\frac{2l}{n(n^2+1)}$
- One end of a thin uniform rod of length L and mass M_1 is rivetted to the centre of a uniform circular disc of radius 'r' and mass M_2 so that both are coplanar. The centre of mass of the combination from the centre of the disc is (Assume that the point of attachment is at the origin)
 1) $\frac{L(M_1 + M_2)}{2M_1}$ 2) $\frac{LM_1}{2(M_1 + M_2)}$ 3) $\frac{2(M_1 + M_2)}{LM_1}$ 4) $\frac{2LM_1}{(M_1 + M_2)}$
- Three particles of masses 1kg, 2kg and 3kg are placed at the vertices A, B and C of an equilateral triangle ABC. If A and B lie at (0,0) and (1,0) m, the co-ordinates of their centre of mass are
 1) $\left(\frac{\sqrt{3}}{2} \text{ m and } \frac{7}{6} \text{ m}\right)$ 2) $\left(\frac{7}{6} \text{ m and } \frac{\sqrt{3}}{4} \text{ m}\right)$ 3) $\left(\frac{7}{12} \text{ m and } \frac{\sqrt{3}}{4} \text{ m}\right)$ 4) $\left(\frac{7}{12} \text{ m and } \frac{7}{12} \text{ m}\right)$

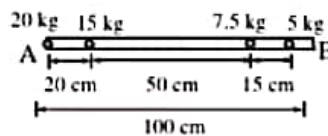
9. Particles of masses 1 kg and 3 kg are at $(2\mathbf{i} + 5\mathbf{j} + 13\mathbf{k})$ m and $(-6\mathbf{i} + 4\mathbf{j} - 2\mathbf{k})$ m then instantaneous position of their centre of mass is

- 1) $\frac{1}{4}(-16\mathbf{i} + 17\mathbf{j} + 7\mathbf{k})$ m 2) $\frac{1}{4}(-8\mathbf{i} + 17\mathbf{j} + 7\mathbf{k})$ m
3) $\frac{1}{4}(-6\mathbf{i} + 17\mathbf{j} + 7\mathbf{k})$ m 4) $\frac{1}{4}(-6\mathbf{i} + 17\mathbf{j} + 5\mathbf{k})$ m

10. Two uniform rods A and B of lengths 5 m and 3 m are placed end to end. If their linear densities are 3 kg/m and 2 kg/m, the position of their centre of mass from their interface is

- 1) 19/14 m on the side of heavier rod 2) 8/7 m on the side of lighter rod
3) 2 m on the side of heavier rod 4) 2 m on the side of lighter rod.

11. For the figure shown, a rod of mass 10 kg (of length 100 cm) with some point masses tied to it at different positions. Find the distance of the point (from A) at which if the rod is picked over a knife edge, it will be in equilibrium about that knife edge.



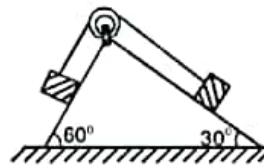
- 1) 26.32 cm 2) 30.43 cm 3) 28.72 cm 4) 32.50 cm

12. Two particles each of the same mass move due north and due east respectively with the same velocity 'V'. The magnitude and direction of the velocity of centre of mass is

- 1) $\frac{V}{\sqrt{2}}$ NE 2) $\sqrt{2} V$ NE 3) $2 V$ SW 4) $\frac{V}{2}$ SW

13. Two blocks of equal mass are tied with a light string which passes over a massless pulley as shown in figure. The magnitude of acceleration of centre of mass of both the blocks is (neglect friction everywhere)

- 1) $\frac{\sqrt{3}-1}{4\sqrt{2}} g$ 2) $(\sqrt{3}-1)g$
3) $\frac{g}{2}$ 4) $\left(\frac{\sqrt{3}-1}{\sqrt{2}}\right)g$



14. Two particles of same mass are projected simultaneously with same speed 20 ms^{-1} from the top of a tower of height 20 m. One is projected vertically upwards and other projected horizontally. The maximum height attained by centre of mass from the ground will be ($g = 10 \text{ ms}^{-2}$)

- 1) $10\sqrt{2} \text{ m}$ 2) 25 m 3) $25\sqrt{2} \text{ m}$ 4) 5 m

15. A balloon of mass M is stationary in air. It has a ladder on which a man of mass m is standing. If the man starts climbing up the ladder with a velocity v relative to ladder, the velocity of balloon is

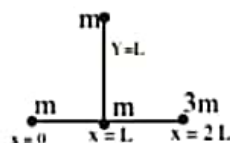
- 1) $\frac{m}{M} v$ upwards 2) $\frac{mv}{(M+m)}$ downwards
3) $\frac{mv}{(M+m)}$ upwards 4) $\frac{Mv}{(M+m)}$ downwards

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. The co-ordinates of the centre of mass of the system as shown in figure are

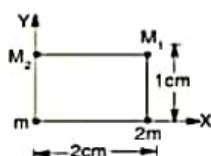
- a) (0, 0)
b) (1.33L, 0.167 L)
c) (L, L)
d) (0.33L, 1.67 L)



2. Four particles, each of mass 1kg, are placed at the corners of a square OABC of side 1m. 'O' is at the origin of the coordinate system. OA and OC are aligned along positive x-axis and positive y-axis respectively. The position vector of the centre of mass is (in 'm')

- a) $\hat{i} + \hat{j}$ b) $\frac{1}{2}(\hat{i} + \hat{j})$ c) $(\hat{i} - \hat{j})$ d) $\frac{1}{2}(\hat{i} - \hat{j})$

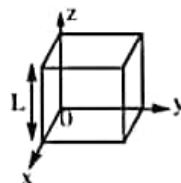
3. Four point masses m , $2m$, M_1 and M_2 are placed on a plane as shown in figure. If the co-ordinates of the centre of mass of the system be $\left(\frac{6}{5}, \frac{7}{10}\right)$ then the values of M_1 and M_2 are respectively



- a) 3m and 4m b) 4m and 3m c) $\frac{29}{4}m$ and $\frac{43}{4}m$ d) $\frac{29}{4}m$ and 6m
4. Two blocks of masses 10kg and 4kg are connected by a spring of negligible mass and placed on a frictionless horizontal surface. An impulse gives a velocity of 14ms^{-1} to the heavier block in the direction of the lighter block. The velocity of the centre of mass is
- a) 30ms^{-1} b) 20ms^{-1} c) 10ms^{-1} d) 5ms^{-1}
5. A shell of mass m in flight explodes into three equal fragments. One of the fragment reaches the ground earlier than the other fragments. The acceleration of the centre of mass of the shell system just after first one reaches ground is
- a) g b) $2g/3$ c) $3g/4$ d) $g/3$
6. A uniform rod AB of mass 2kg and length 4m is kept vertically on the smooth horizontal floor with its end A in contact with floor. On slightly disturbing rod, it comes into motion such that end A is always maintaining contact with smooth horizontal floor. Then distance (in m) travelled by centre of mass of rod during time interval in which its other end B strikes the smooth floor will be _____
- a) 2 b) 1 c) 3 d) 4
7. Two particles of masses 2 m and 3 m are at a distance 'd' apart. Under their mutual gravitational force, they start moving towards each other. The acceleration of their centre of mass when they are $d/2$ apart is
- a) $\frac{12 Gm}{d^2}$ b) $\frac{8 Gm}{d^2}$ c) $\frac{24 Gm}{d^2}$ d) zero

8. Figure shows a cubical box that has been constructed from uniform metal plate of negligible thickness. The box is open at the top and has edge length 40 cm. The z co-ordinate of the centre of mass of the box in cm, is :

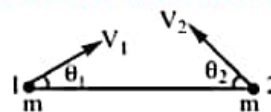
- a) 12
b) 16
c) 20
d) 22



9. Two balls of masses $2M$ and $6M$ have radii $2R$ and $3R$. Their centre of masses are separated by $10R$. They move towards each other under their gravitational force. The distance moved by the centre of smaller sphere when the spheres touch each other is
a) $5/4R$ b) $7/2R$ c) $10/3R$ d) $15/4R$
10. A shell projected from a level ground has a range R , if it did not explode. At the highest point, the shell explodes into two fragments having masses in the ratio $1:2$; with each fragment moving horizontally immediately after the explosion. If the lighter fragment falls at a distance $R/2$ from the point of projection, behind the point of projection, the distance at which the other fragments falls from the point of projection is
a) $R/2$ b) $2R/3$ c) $4R/3$ d) $7R/4$
11. A shell is projected from a level ground with a velocity u at angle θ the horizontal. When the shell is at the highest point it explodes into two equal fragments. One of the fragments retraces its path to the point of projection. At what distance from the point of projection does the other fragment fall.
a) $\frac{2u^2 \sin 2\theta}{g}$ b) $\frac{u^2 \sin 2\theta}{g}$ c) $\frac{u^2 \sin 2\theta}{2g}$ d) $\frac{2u^2 \sin 2\theta}{3g}$

More than One correct answer Type Questions

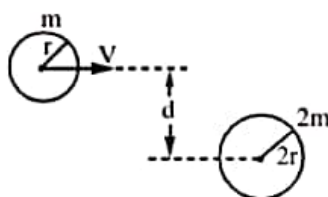
12. Which of the following is/are correct ?
a) If centre of mass of three particles is at rest and it is known that two of them are moving along different lines then the third particle must also be moving
b) If centre of mass remains at rest, then net workdone by the forces acting on the system must be zero
c) If centre of mass remains at rest, then the net external force must be zero
d) If speed of centre of mass is changing, then there must be some net work being done on the system from outside
13. Two particles of equal mass ' m ' are projected from the ground with speed V_1 and V_2 at angles θ_1 and θ_2 as shown in fig. Given $\theta_2 > \theta_1$ and $V_1 \cos \theta_1 = V_2 \cos \theta_2$ which statements is/are correct ?
a) Centre of mass of particles will move along a vertical line
b) Centre of mass of particles will move along a line inclined at some angle with vertical
c) Particle '1' will be above centre of mass level when both particles are in air
d) Particle '2' will be above centre of mass level when both particles are in air



Linked Comprehension Type Questions

Passage-I :

A uniform sphere with mass $2m$ and radius $2r$ is at rest in a place where no external forces are applied. A second sphere of mass m and radius r moves towards it with a velocity v (without rotating). The two spheres stick together and then continue their motion as one body. Given that $d = 2r$



14. The distance of the centre of mass of the system from the centre of bigger sphere at the moment of collision is
- a) r b) $2r$ c) $3r$ d) $4r$
15. The velocity of centre of mass immediately after the collision is
- a) $\frac{2V}{3}$ b) $\frac{2V}{5}$ c) $\frac{3V}{2}$ d) $\frac{V}{3}$
16. The moment of inertia about the centre of mass of the unified body after the collision is
- a) $\frac{24}{5}mr^2$ b) $\frac{24}{3}mr^2$ c) $\frac{48}{5}mr^2$ d) $\frac{48}{3}mr^2$

Matrix Matching Type Questions

17. Centre of mass of a system is a unique point which is very helpful in solving the problems related to the motion of rigid bodies. In columns I and II, some statements regarding centre of mass are given. Match the Columns :

Column-I

- A) In the absence of external forces, internal forces
B) Centre of mass of a body (can be)
C) C.M. of a ring
D) Centre of mass of solid cylinder is

Column-II

- P) Inside the material of the the body.
Q) May affect the motion of centre of mass
R) Does not affect the motion of centre of mass
S) Outside the material of the body

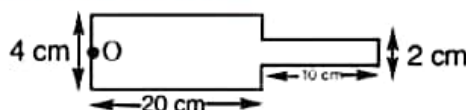
EXERCISE-II

(Method of Symmetry, Method of Decomposition, Method of Integration, Centre of Mass with Spring Forces)

LEVEL-I (MAIN)

Straight Objective Type Questions

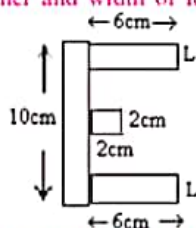
1. A rod consists of two parts made up of a homogeneous material with square type cross – section as shown. The position of C.M from the end 'O' is



- 1) $\frac{35}{3}$ cm 2) $\frac{65}{3}$ cm 3) $\frac{55}{3}$ cm 4) 15 cm

2. The co-ordinates of centre of mass of letter E which is cut from a uniform metal sheet are (Take origin at bottom left corner and width of letter 2 cm every where)

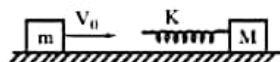
- 1) (2cm, 4cm)
2) (2.4cm, 5cm)
3) (3cm, 5cm)
4) 3.3cm, 5cm)



3. The centre of mass of a non uniform rod of length L whose mass per unit length $\lambda = \frac{Kx^2}{L}$, where K is a constant and x is the distance from one end is

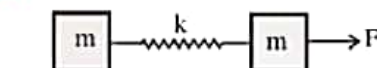
- 1) $3L/4$ 2) $L/8$ 3) K/L 4) $3K/L$

4. A block of mass m moves with a velocity v_0 towards a stationary block of mass M (connected to a spring) on a smooth horizontal surface (see figure). If spring constant is k , what is the maximum compression in the spring?



- 1) $v_0 \sqrt{\frac{mM}{2k(m+M)}}$ 2) $v_0 \sqrt{\frac{mM}{k(m+M)}}$ 3) $v_0 \sqrt{\frac{2mM}{k(m+M)}}$ 4) None of these

5. Two blocks of equal mass m are connected by an unstretched spring and the system is kept at rest on a frictionless horizontal surface. A constant force F is applied on the first block pulling it away from the other as shown in figure.

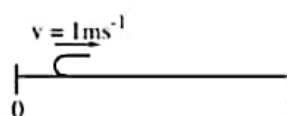


If the extension of the spring is x_0 at time t , then the displacement of the first block at this instant is ____

- 1) $\frac{1}{2} \left(\frac{Ft^2}{2m} + x_0 \right)$ 2) $\frac{1}{2} \left(\frac{Ft^2}{2m} + x_0 \right)$ 3) $\frac{1}{2} \left(\frac{Ft^2}{2m} - x_0 \right)$ 4) $\left(\frac{Ft^2}{2m} + x_0 \right)$

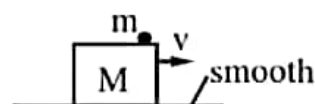
6. A long, thin pliable carpet is laid on the floor. One end of the carpet is bent back and then pulled backwards with constant velocity $v = 1 \text{ ms}^{-1}$ just above the part of the carpet which is still at rest on the floor. Length of the carpet is 1 meter. Then velocity of centre of mass of the moving part is

- 1) 1 ms^{-1} 2) $\frac{3}{4} \text{ ms}^{-1}$ 3) $\frac{1}{2} \text{ ms}^{-1}$ 4) $\frac{1}{4} \text{ ms}^{-1}$



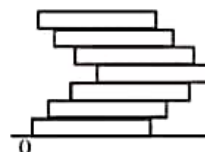
7. A particle of mass m is kept at the edge of a smooth cube of mass M and side L as shown in figure. Cube starts moving at $t = 0$ with a constant velocity v . Displacement of the center of mass along the horizontal direction from $t = 0$ to when particle hits the ground is

- 1) $\frac{mv}{m+M} \left[\frac{L}{v} + \sqrt{\frac{2L}{g}} \right]$ 2) $\frac{mv}{m+M} \left[\frac{v}{L} + \sqrt{\frac{2L}{g}} \right]$
3) $\frac{Mv}{m+M} \left[\frac{L}{v} + \sqrt{\frac{2L}{g}} \right]$ 4) $\frac{Mv}{m+M} \left[\frac{v}{L} + \sqrt{\frac{2L}{g}} \right]$



8. Seven homogeneous blocks each of length L are arranged as shown in the figure. Each block is displaced with respect to the one in contact by $L/10$. The x -coordinate of the centre of mass of the system relative to the origin O is

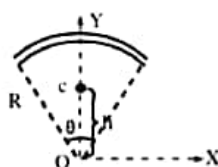
- 1) $\frac{11L}{12}$ 2) $\frac{22L}{25}$
3) $\frac{22L}{35}$ 4) $\frac{11L}{35}$



LEVEL-II (ADVANCED)

Straight Objective Type Questions

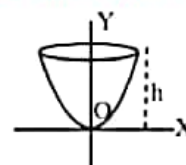
1. From the figure there is an arc of a thin ring, that subtends an angle ' θ ' at the centre. The mass of the arc is m and its radius is R . Locate the centre of mass of the arc from ' O ' to c is ' h ' then



- a) $h = \frac{2R \sin(\theta/2)}{\theta}$ b) $h = \frac{\sin(\theta/2)}{\theta}$ c) $h = \frac{\sin \theta}{\theta}$ d) $h = \frac{R \sin(\theta/2)}{\theta}$

2. A paraboloid shaped solid object is formed by rotating on parabola $y = 2x^2$ about y -axis as shown in figure. If the height of the body is ' h ' then the position of centre of mass from origin (Assume density to be uniform throughout) is

- a) $\frac{h}{3}$ b) $\frac{h}{4}$
c) $\frac{2h}{3}$ d) $\frac{2h}{4}$



3. A small particle of mass m starts sliding down from rest along the smooth surface of a fixed hollow hemisphere of mass M ($4m$). The distance of centre of mass of (particle + hemisphere) from centre O of hemisphere, when the particle separates from the surface of hemisphere is $\frac{\sqrt{69}}{5\alpha}R$. Find the value of ' α '



- a) 1 b) 2 c) 3 d) 4

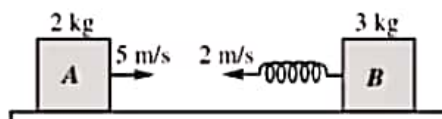
More than One correct answer Type Questions

4. Which of the following statement is/are correct about location of centre of mass?
- The centre of mass of three particles of equal weight placed at the vertices of a triangle ABC coincides with the centroid of triangle ABC.
 - A uniform wire is bent to form the sides of a triangle ABC. The centre of mass of wire coincides with centroid of triangle ABC.
 - The centre of mass of four particles of equal weight placed at the vertices of a quadrilateral coincides with centroid of quadrilateral.
 - The centre of mass of a body can exist outside the material of the body.

Linked Comprehension Type Questions

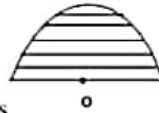
Passage - I :

In Fig. Block A of mass 2kg is moving to the right with a speed 5m/s on a horizontal frictionless surface. Another block B of mass 3kg, with a massless spring attached to it, is moving to the left on the same surface and with a speed 2m/s. Let us take the direction to the right as the positive x-direction. At some instant, block A collides with the spring attached to block B. At some other instant, the spring has maximum compression and then, finally, the blocks move with their final velocities. Assuming that (i) the spring force is conservative and so there is no conversion of kinetic energy to internal energy and (ii) no sound is made when block A hits the spring.



5. Final velocity of block A will be
 a) $2.5\hat{i}\text{ m/s}$ b) $-1.8\hat{i}\text{ m/s}$ c) $3.6\hat{i}\text{ m/s}$ d) $-3.4\hat{i}\text{ m/s}$
6. Final velocity of centre of mass of the system of blocks A and B will be
 a) Zero b) $0.6\hat{i}\text{ m/s}$ c) $0.8\hat{i}\text{ m/s}$ d) $-1.4\hat{i}\text{ m/s}$

Integer Type Questions

7. A rod AB of l is such that its linear density (mass per unit length) μ varies as $\mu = \frac{a}{b-x}$, where x is the distance of the section from end A (and $b > l$). The distance of the centre of mass from end A is $b - \frac{zl}{\ln\left(\frac{b}{b-l}\right)}$ where z is
8. A semi-circular plate of radius R has density $\rho = \rho_0 r$, where r is the distance from centre. The position of centre of mass from O is $\frac{1.5yR}{2\pi}$ where y is 
9. The mass per unit length of a non-uniform rod of length L varies as $m = m_0 \frac{x^2}{L}$, where m_0 is a constant and x is the distance of any point on the rod measured from one end. The centre of mass of the rod is at $\frac{PL}{4}$ where P is

EXERCISE-III

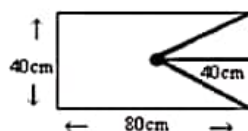
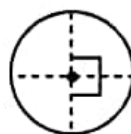
(Shift in Centre of Mass)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. A uniform disc of radius R is put over another uniform disc of radius $2R$ of same thickness and density. The peripheres of the two discs touch each other. The position of their centre of mass is
 - 1) at $R/3$ from the centre of the bigger disc towards the centre of the smaller disc
 - 2) at $R/5$ from the centre of the bigger disc towards the centre of the smaller disc
 - 3) at $2R/5$ from the centre of the bigger disc towards the centre of the smaller disc
 - 4) none of the above

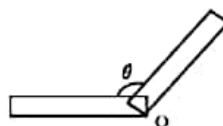
2. Six identical particles each of mass m are arranged at the corners of a regular hexagon of side length a . If the mass of one of the particle is doubled, the shift in the centre of mass is
- 1) a 2) $\frac{6a}{7}$ 3) $\frac{a}{7}$ 4) $\frac{a}{\sqrt{3}}$
3. A circular plate has a uniform thickness and has a diameter 56 cm. A circular disc of diameter 42 cm is removed from one edge of the plate. The distance of centre of mass of the remaining portion from the centre of the circular plate is
- 1) 18 cm 2) 9 cm 3) 27 cm 4) 4.5 cm
4. If a square of side $R/2$ is removed from a uniform circular disc of radius R as shown in the figure, the shift in centre of mass is
- 1) $\frac{R}{4\pi-1}$ 2) $\frac{R}{2(4\pi-1)}$
- 3) $\frac{R}{3(4\pi-1)}$ 4) $\frac{R}{4(4\pi-1)}$
5. If a triangular sheet is removed from one edge of a uniform rectangular sheet as shown in the figure, then the shift in the centre of mass of the system is
- 1) 3 cm 2) 6 cm
- 3) 9 cm 4) 12 cm
6. From a circle of radius a , an isosceles right angled triangle with the hypotenuse as the diameter of circle is removed. The distance of the centre of gravity of the remaining portion from the centre of the circle is:
- 1) $3(\pi-1)a$ 2) $\frac{(\pi-1)a}{6}$ 3) $\frac{a}{3(\pi-1)}$ 4) $\frac{a}{3(\pi+1)}$
7. A boat of mass 40 kg is floating on water. A boy of mass 10 kg on the boat moves by 8 m towards the shore. The distance by which the boat moves away from the shore is
- 1) 4 m 2) 3 m 3) 2 m 4) 1.6 m



LEVEL-II (ADVANCED)

Straight Objective Type Questions

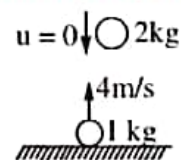
1. A uniform disc of radius $R/2$ is put over another uniform disc of radius R of same thickness and density. The peripheries of the two discs touch each other. The centre of mass of the system from centre of big disc is
- a) $R/10$ b) $R/20$ c) $9R/10$ d) $11R/10$
2. A thin uniform rod of length L is bent at its mid point as shown in the figure. The distance of centre of mass from the point 'O' is
- a) $\frac{L}{2} \cos \frac{\theta}{2}$ b) $\frac{L}{4} \sin \frac{\theta}{2}$
- c) $\frac{L}{4} \cos \frac{\theta}{2}$ d) $\frac{L}{2} \sin \frac{\theta}{2}$



3. A uniform wire of length L is bent in the form of a circle. The shift in its centre of mass is
- a) $\frac{L}{\pi}$ b) $\frac{2L}{\pi}$ c) $\frac{L}{2\pi}$ d) $\frac{L}{3\pi}$
4. Four identical particles each of mass 1 kg are arranged at the corners of a square of side length $2\sqrt{2} \text{ m}$. If one of the particles is removed, the shift in the centre of mass is
- a) $\frac{8}{3} \text{ m}$ b) $\frac{4}{3} \text{ m}$ c) $\frac{2}{3} \text{ m}$ d) 2 m

More than One correct answer Type Questions

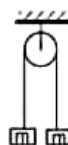
5. A disc of radius ' r ' is removed from the disc of radius ' R ' then
- a) The minimum shift in centre of mass is zero
- b) The maximum shift in centre of mass cannot be greater than $\frac{r^2}{(R+r)}$
- c) Centre of mass must lie where mass exists
- d) The shift in centre of mass is $\frac{r^2}{(R+r)}$
6. A ball of mass 1 kg is thrown up with an initial speed of 4 m/s . A second ball of mass 2 kg is released from rest from some height as shown in the figure
- a) The centre of mass first moves up and down with acceleration $g/3$
- b) The centre of mass first moves up and then comes down
- c) The acceleration of the centre of mass is g downwards
- d) The centre of mass of the two balls remains stationary



Matrix Matching Type Questions

7. Column-I

- A) The block plus wedge system is placed over smooth horizontal surface. After the system is released from rest, the centre of mass of system
- B) The string connecting both the blocks of mass m is horizontal. Left block is placed over smooth horizontal table as shown. After the two block system is released from rest, the centre of mass of system
- C) The block and monkey have same mass. The monkey starts climbing up the rope. After the monkey starts climbing up, the centre of mass of monkey + block system
- D) Both blocks of mass m are initially at rest. The left block is given initially velocity u Downwards. Then, the centre of mass of two block system afterwards.

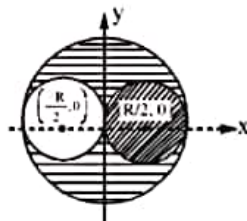


Column-II

- P) Shift towards right
- Q) Shifts downwards
- R) Shifts upwards
- S) Does not shift

Integer Type Questions

8. Figure shows a uniform disc of radius R from which a hole of radius $R/2$ has been cut out from left of the centre and is placed on right of the centre of disc. The centre of mass of resulting disc is at a distance $R/2x$ from centre of uniform disc. Then x is



KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL - I

- 1) 3 2) 1 3) 2 4) 3 5) 2 6) 1 7) 2 8) 3
9) 1 10) 1 11) 2 12) 1 13) 1 14) 2 15) 2

LEVEL - II

- 1) b 2) b 3) a 4) c 5) b 6) a 7) d 8) b
9) d 10) d 11) a 12) ac 13) ad 14) a 15) d 16) c
17) A-R; B-PS; C-S; D-P

EXERCISE-II

LEVEL - I

- 1) 1 2) 4 3) 1 4) 2 5) 1 6) 2 7) 3 8) 3

LEVEL - II

- 1) a 2) c 3) c 4) ad 5) d 6) c 7) 1 8) 2
9) 3

EXERCISE-III

LEVEL - I

- 1) 2 2) 3 3) 2 4) 4 5) 3 6) 3 7) 4

LEVEL - II

- 1) a 2) c 3) c 4) c 5) ab 6) abc
7) A-Q; B-PQ; C-R; D-S 8) 2